

Aspect-Based Sentiment Analysis of After-Sales Service Quality: A Case Study of Snowa and Competitors Using Digikala Reviews

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Abstract—In recent years, aspect-based sentiment analysis has gained attention as a method for processing and summarizing customer reviews. This type of analysis aims to answer two main questions: First, which aspect of the product is the review referring to? Second, what is the customer's sentiment towards that aspect? Given the importance of after-sales service quality in customer satisfaction, this study examines an aspect-based analysis of a set of reviews from Digikala regarding the after-sales service quality of three washing machine brands, including Snowa and its two competitors. Deep learning models are implemented and evaluated based on example-based and label-based metrics. At the end, an analysis is conducted on customer satisfaction with the quality of after-sales services for the three selected brands.

Keywords— *After-Sales Service, Aspect-Based Sentiment Analysis, Deep Learning, Multi-Label Classification*

I. INTRODUCTION

Customer satisfaction is a fundamental factor in the growth of businesses. Companies that fail to satisfy their customers will eventually be eliminated from the competitive market. Customer satisfaction is influenced by the features of the product or service that encourage the customer to buy and use it. On the other hand, dissatisfaction arises from deficiencies or shortcomings in the product or service, leading to customer complaints and discontent [1]. Customer satisfaction with products and services strengthens customer loyalty and helps retain customers. Also, satisfied customers are likely to tell others about their good experience; This is a free advertisement for the company that may attract new customers. This is especially important in Eastern cultures, where social life is formed in a way that improves social communication with others in society [2].

One effective way to achieve customer satisfaction is to identify their needs and take necessary actions to meet those needs, as well as to identify strengths and weaknesses in services and address the causes of dissatisfaction. Hence, understanding customer needs and fulfilling them is one of the

most important tasks for any business [3]. One way to understand customer needs is by receiving and analyzing their feedback. Collecting and acting on customer suggestions increases satisfaction.

In recent years, especially after the COVID-19 epidemic, many businesses have gone to sell their products on the Internet. These businesses introduce and sell their products on their own websites, e-commerce platforms, or social media pages. Many customers share their experiences using the products on these platforms. The posts and comments on these networks, as well as ratings and reviews posted under each product on sales websites, can serve as a source for understanding customer opinions and needs. These reviews reflect customer satisfaction or dissatisfaction regarding their experience with the product or service [4]. Therefore, analyzing online customer reviews is one of the key strategies for understanding customer needs.

To understand customer needs through online review analysis, the first step is to gather customer reviews. The next step is to extract useful and necessary information from each review using text mining techniques and natural language processing algorithms. Due to the rapid increase in online reviews across social media, it is necessary to process unstructured or semi-structured textual data using text mining technology [5]. Text mining involves modifying, structuring, and examining vast collections of documents to extract relevant information and identify connections between different elements [6].

One of the key factors influencing customer satisfaction during the purchase and usage of a product is the quality of after-sales services [7]. Services such as installation scheduling, technical expert expertise, and follow-up problems during the installation process and then using the product have a great impact on customer satisfaction.

Considering the importance of customer needs and opinions, as well as the role of after-sales service quality as a key factor in customer satisfaction, this study aims to examine